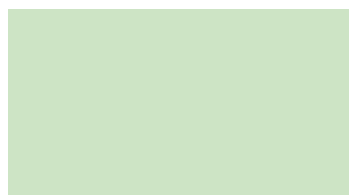
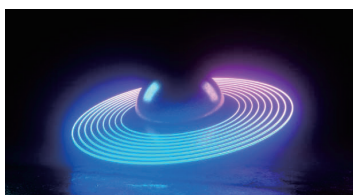
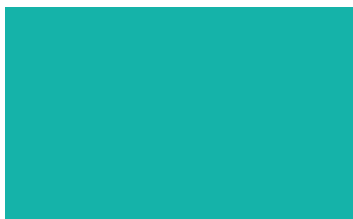
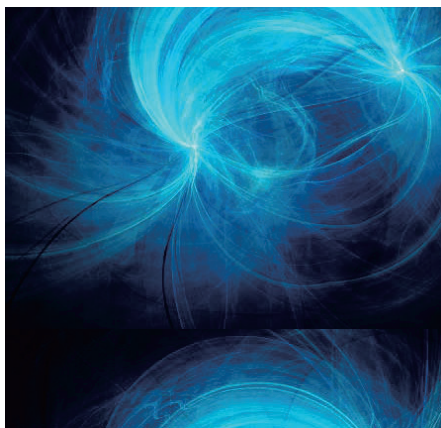


TestDaily AP Course

AP Physics C: Electricity and Magnetism AP Course



Unit 1: Electrostatics

AP Physics C Electricity and Magnetism



Contents

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1

01 Charge and Coulomb's Law

02 Electric Field

03 Gauss's Law

04 Electric Potential

TESTDAILY

Charge and Coulomb's Law

1.1

TESTDAILY

Charge and Coulomb's Law

1.1.1 — Charge

1.1.4 — Superposition

1.1.2 — Polarization and Induced Charges

1.1.5 — Motion

1.1.3 — Coulomb's Law

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1.1 Charge and Coulomb's Law

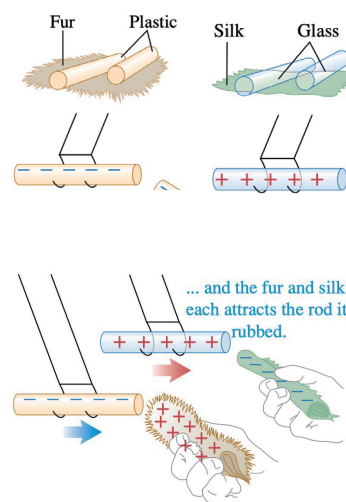
1.1.1 Charge



Property

Electric Charge

- There are two kinds of charges: **positive** and **negative**
 - Glass rod rubbed with silk is positively charged
 - Plastic rod rubbed with fur is negatively charged
- The Law of Electrostatics states that **like** charges **repel** and **unlike** charges **attract** through electrostatic interactions. (CED p31)

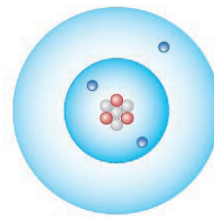


Property

TESTDAILY

Electric Charge

- The structure of matter
Matter — molecules — atoms —
electrons and atomic nucleus (neutrons and protons)
 - electrons are negatively charged
 - protons are positively charged
 - neutrons are neutral
 - atoms are neutral
- Elementary charge e



(a) Neutral lithium atom (Li):
 3 protons (3+)
 4 neutrons
 3 electrons (3-)
 Electrons equal protons:
 Zero net charge

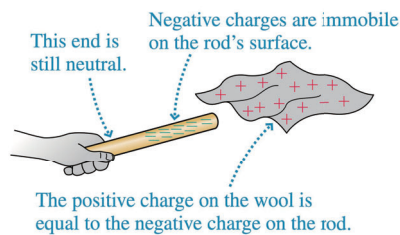
1.1.1

Theorem

TESTDAILY

Principle of Conservation of Charge (Young 13e p690)

The algebraic **sum** of all the electric charges in any **closed** system is constant.



1.1.1

1.1 Charge and Coulomb's Law

TESTDAILY

1.1.2 Polarization and Induced Charges

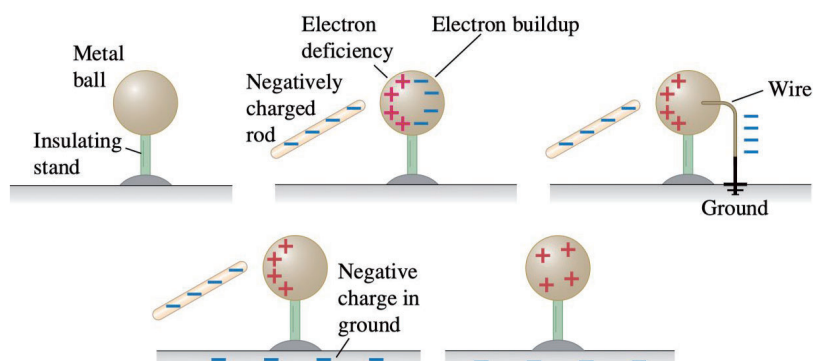
1.1.2

Property

TESTDAILY

Charging by Induction

- Bringing a charged body close to a **neutral** object can polarize the object by creating **induced charges** on the surface of the object.
- In conductors, charge can move freely.



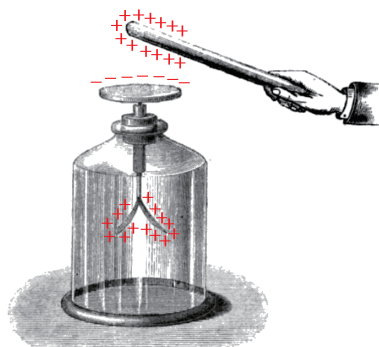
1.1.2

Property

TESTDAILY

Charging by Induction

- Electroscope: detect the **presence** of electric charge on a body.



1.1.2

Property

TESTDAILY

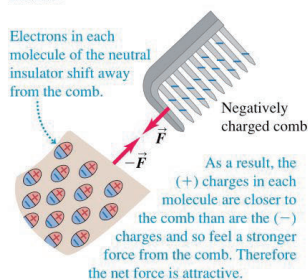
Charging by Induction

- In insulators, polarization occurs **within** the molecules.

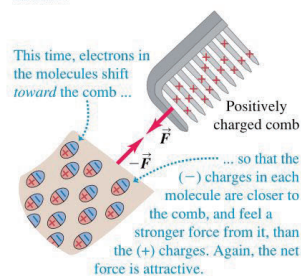
(a) A charged comb picking up uncharged pieces of plastic



(b) How a negatively charged comb attracts an insulator



(c) How a positively charged comb attracts an insulator



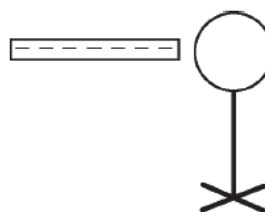
1.1.2

Exercise

2015_Intl_MCQ_10

A spherical conductor is on an insulating stand, as shown in the figure above. A negatively charged rod is brought close to the sphere but does not touch the sphere. Which of the following describes the resulting charge on the sphere?

- (A) Positive
- (B) Negative
- (C) No net charge, but the sphere is polarized with positive charge on the left side.
- (D) No net charge, but the sphere is polarized with negative charge on the left side.
- (E) No net charge and no polarization



1.1.2

1.1 Charge and Coulomb's Law

1.1.3 Coulomb's law

1.1.3

Theorem

Coulomb's Law (Young 13e p694, CED p32)

- The magnitude of the electric force or **electrostatic** force between two **point** charges q_1 and q_2 , separated by a distance r , is given by

$$|\vec{F}_E| = k \left| \frac{q_1 q_2}{r^2} \right|$$

- Electric Charge q 电荷量

- Unit: coulomb, C 库伦
- Elementary charge e

$$e = 1.602176634 \times 10^{-19} \text{ C} \approx 1.6 \times 10^{-19} \text{ C}$$

- Coulomb's constant k and vacuum permittivity ϵ_0

$$k = \frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)$$

1.1.3

Discussion

TESTDAILY

Coulomb's Law

- Compare with the gravitational force

$$F_G = G \frac{m_1 m_2}{r^2} \quad F_E = k \frac{q_1 q_2}{r^2}$$

- What are the differences and similarities? Magnitude/direction/...



Example

TESTDAILY

Princeton_2020_Chapter12_Example1

Consider two small spheres, one carrying a charge of +1.5 nC and the other a charge of -2.0 nC, separated by a distance of 1.5 cm. Find the electric force between them.

Answer: 1.2×10^{-4} N

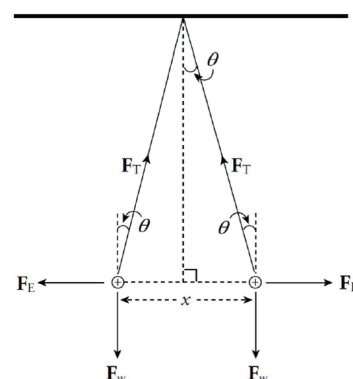


Example

TESTDAILY

Princeton_2020_Chapter12_Example4

Two pith balls of mass m are each given a charge of $+q$. They are hung side-by-side from two threads each of length L and move apart as a result of their electrical repulsion. Find the equilibrium separation distance x in terms of m , q , and L . (Use the fact that if θ is small, then $\tan \theta \approx \sin \theta$.)



Answer: $x = \sqrt[3]{\frac{2kq^2L}{mg}}$



1.1 Charge and Coulomb's Law

1.1.4 Superposition



Method

TESTDAILY

Superposition

- The net electric force acting on a point charge is simply the **vector** sum of all electric forces due to other point charges.

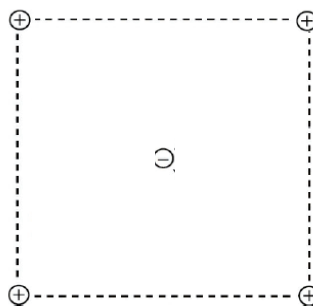


Example

TESTDAILY

Princeton_2020_Chapter12_Example2&3

- (1) Consider four equal, positive point charges that are situated at the vertices of a square. Find the net electric force on a negative point charge placed at the square's center.
- (2) If the two positive charges on the bottom side of the square in the previous example were removed, what would be the net electric force on the negative charge?
- Assume that each side of the square is s , each positive charge is $+Q$, and the negative charge is $-q$.



Answer: (1) 0; (2) $2\sqrt{2}k \frac{Qq}{s^2}$.



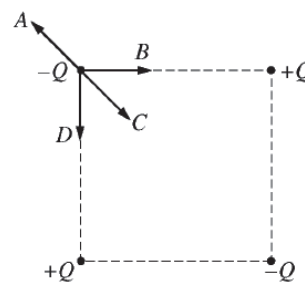
Exercise

TESTDAILY

2016_Intl_MCQ_35

Four point charges of equal magnitude but different signs are arranged on the corners of a square as shown above. Which of the vectors shown represents the direction of the net force acting on the charge at the upper left-hand corner of the square due to the other charges?

- (A) *A*
 (B) *B*
 (C) *C*
 (D) *D*
 (E) It has no direction because the force is zero.



1.1 Charge and Coulomb's Law

1.1.5 Motion



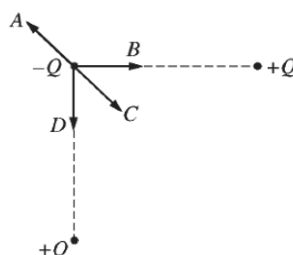
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Example

TESTDAILY

Motion Under the Influence of Electric Force

- (a) Determine the initial acceleration.
 (b) Describe the subsequent motion.



Answer: (a) $a = \frac{\sqrt{2}kQ^2}{s^2m}$.



Electric Field



TESTDAILY

Electric Field

1.2.1 — Electric Field of a Point Charge

1.2.2 — Superposition of Electric Fields

1.2.3 — Electric Fields of Different Charge Distributions

TESTDAILY

1.2 Electric Field

1.2.1 Electric Field of a Point Charge



TESTDAILY

Introduction

TESTDAILY

Electric Field

- When two electrically charged particles in empty space interact, how does each one know the other is there?
- Think about gravitational field.

$$\vec{g} = -\frac{GM}{r^2} \hat{r}, \quad \vec{F}_g = m\vec{g}$$

Two-stage process:

- Object A modifies the properties of the space around it.
 - Object B responds by experiencing a force.
- The electric force on a charged body is exerted by the **field** created by other charged bodies.



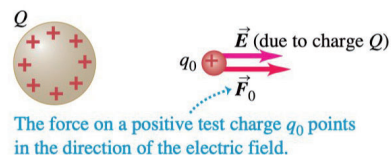
Definition

TESTDAILY

Electric Field

- The electric field at a point is equal to the electric force experienced by a positive **test charge** q_0 at that point, divided by the charge q_0 . The test charge has negligible size and mass. (CED p33 & Young 13e p699)
- Electric field is a vector with both magnitude and direction

Physical quantity	Electric field 电场
Definition	The electric force per unit charge
Symbol	$\vec{E}$
Equation	$\vec{E} = \frac{\vec{F}}{q_0}$
Unit	newton per coulomb, N/C



Property

TESTDAILY

Electric Field of a Point Charge

- Direction
- Magnitude

$$E = \frac{F_0}{q_0} = \frac{k \frac{qq_0}{r^2}}{q_0} = k \frac{q}{r^2}$$



Definition

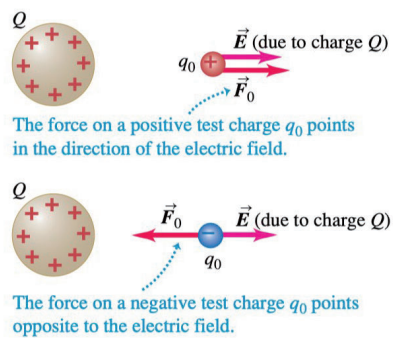
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Electric Field

- What if the test charge is negative?

$$\vec{E} = \frac{\vec{F}}{q_0}$$

Field is determined only by the source, regardless of the test charge!



1.2.1

Property

TESTDAILY

Predict Force From Field

$$\vec{F} = q\vec{E}$$

The force experienced by a point charge q :

- positive
- negative

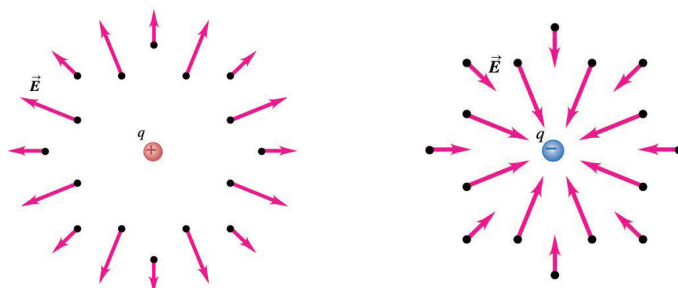
1.2.1

Concept

TESTDAILY

Vector Field

- An infinite set of vector quantities, one associated with each point in space.



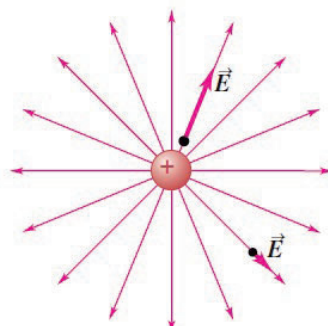
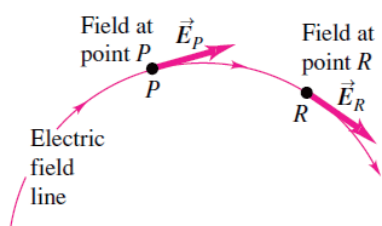
1.2.1

Concept

TESTDAILY

Electric Field Lines

- Its tangent at any point is in the direction of the electric-field vector at that point.
- Its density describes the strength.



1.2.1

1.2 Electric Field

1.2.2 Superposition of Electric Fields

1.2.2

Method

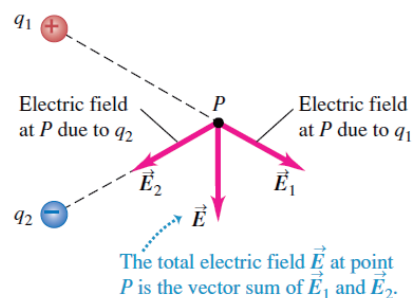
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Superposition of Electric Fields

- The total electric field is the **vector sum** of the fields due to each point charge in the charge distribution.

$$\vec{F}_0 = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots = q_0 \vec{E}_1 + q_0 \vec{E}_2 + q_0 \vec{E}_3 + \dots$$

$$\vec{E} = \frac{\vec{F}_0}{q_0} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$$

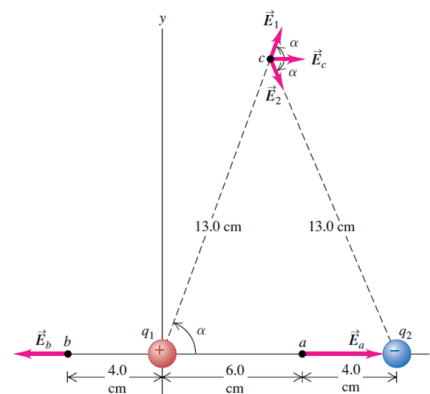


1.2.2

Example

Young_13e_p704_Example21.8

Point charges $q_1 = +12 \text{ nC}$ and $q_2 = -12 \text{ nC}$ are 0.100 m apart. (Such pairs of point charges with equal magnitude and opposite sign are called **electric dipoles**.) Compute the electric field caused by q_1 the field caused by q_2 , and the total field (a) at point a ; (b) at point b ; and (c) at point c .



Answer: (a) $9.8 \times 10^4 \text{ N/C}$; (b) $6.2 \times 10^4 \text{ N/C}$; (c) $4.9 \times 10^3 \text{ N/C}$

1.2.2

1.2 Electric Field

1.2.3 Electric Fields Of Different Charge Distributions

1.2.3

Learning Objective

Different Charge Distributions

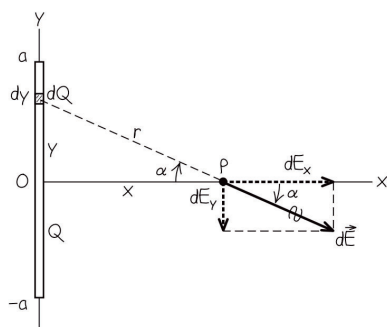
- Uniformly charged straight wire
 - Extension: along the line of uniformly charged rod
- Uniformly charged ring
- Uniformly charged disk
- Uniformly charged semicircle

1.2.3

Example

TESTDAILY

Young_13e_p706_Example21.10: Straight, Uniformly Charged Wire



Positive charge Q is distributed uniformly along the y -axis between $y = -a$ and $y = +a$. Find the electric field at point P on the x -axis at a distance x from the origin.

$$\int (x^2 + y^2)^{-\frac{3}{2}} dy = \frac{y}{x^2 \sqrt{x^2 + y^2}}$$

1.2.3

Example

TESTDAILY

Straight, Uniformly Charged Wire

$$E = \frac{kQ}{x\sqrt{x^2 + a^2}}$$

When a is very small?

When a is very big?

1.2.3

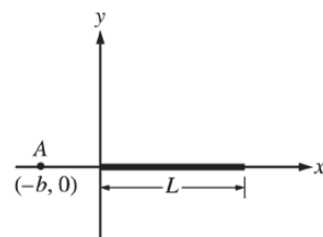
Example

TESTDAILY

2016_Intl_FRQ_1

The rod is replaced with one of the same length but with a uniform positive linear charge density λ_0 .

(c) Show that the electric field at point A , which is a distance b to the left of the rod, has a magnitude of $E = \frac{\lambda_0 L}{4\pi\epsilon_0 b(b+L)}$.



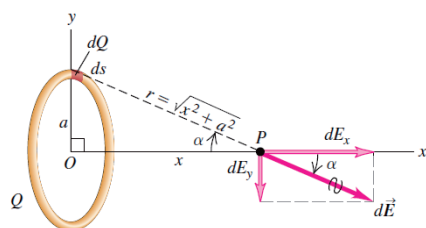
1.2.3

Example

TESTDAILY

Young_13e_p705_Example21.9: Uniformly Charged Ring

Charge Q is uniformly distributed around a conducting ring of radius a . Find the electric field at a point P on the ring axis at a distance x from its center.



1.2.3

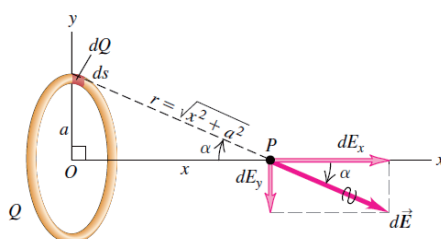
Example

TESTDAILY

Uniformly Charged Ring

$$E = k \frac{Qx}{(x^2 + a^2)^{3/2}}$$

When $x = 0$?



When x is very big?

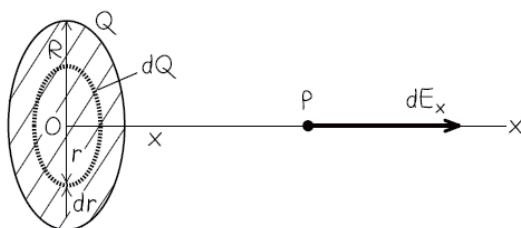
1.2.3

Example

TESTDAILY

Young_13e_p707_Example21.11: Uniformly Charged Disk

A nonconducting disk of radius R has a uniform positive surface charge density σ . Find the electric field at a point along the axis of the disk a distance x from its center. Assume that x is positive.



1.2.3

Example

TESTDAILY

Uniformly Charged Disk

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{1}{\sqrt{(R/x)^2 + 1}} \right]$$

If the disk is very large or the point is very close to the disk?

If the disk is very small or the point is very far from the disk?

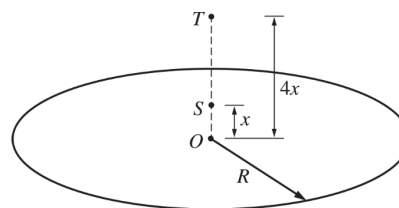
1.2.3

Exercises

TESTDAILY

2016_Intl_MCQ_10

The figure above shows a thin, circular nonconducting sheet of positive charge uniformly distributed over its area. The radius of the sheet is R . Point O is at the center of the sheet. Point S is a distance x from the center of the sheet, and point T is a distance $4x$ from the center of the sheet. Assume $R \gg x$. If the magnitude of the electric field at point T is E_T , which of the following best represents the magnitude of the electric field at point S ?



Note: Figure not drawn to scale.

- (A) $E_T/16$
- (B) $E_T/4$
- (C) E_T
- (D) $4E_T$
- (E) $16E_T$

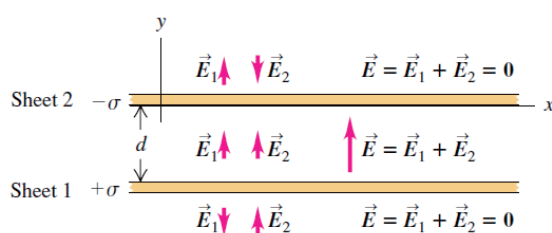
1.2.3

Property

TESTDAILY

Uniform Electric Field

- Infinite plane sheet of charge.
- Magnitude
- Direction



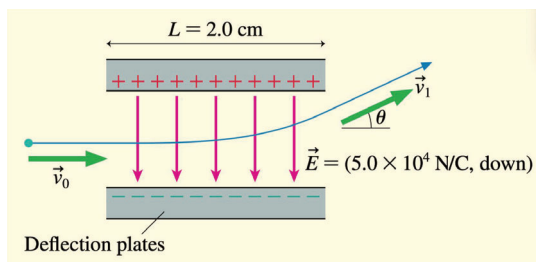
1.2.3

Property

TESTDAILY

Motion in Uniform Electric Field

- Similar to projectile motion



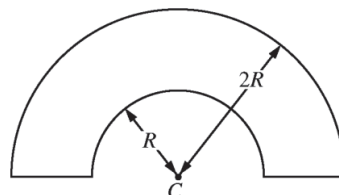
1.2.3

Example

TESTDAILY

2015_Intl_FRQ_3

A thin nonconducting wire is shaped into a loop containing two concentric semicircular arcs with their centers at point C , as shown above. The wire carries a positive uniform linear charge density λ . Express your answers in parts (a) and (b) in terms of R , λ , θ , and fundamental constants, as appropriate.



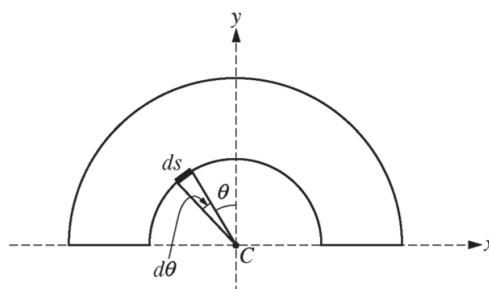
1.2.3

Example

TESTDAILY

2015_Intl_FRQ_3

(a) Derive an expression for the y -component of the infinitesimally small electric field, dE_y , produced at point C by the charge on the small piece of wire in terms of the infinitesimally small angle $d\theta$ shown in the figure on the right.



1.2.3

Example

TESTDAILY

2015_Intl_FRQ_3

(b) Using the expression from part (a), derive an expression for the magnitude of the electric field at point C produced by the entire wire.

**Gauss's Law**

TESTDAILY

Gauss's Law**1.3.1 — Flux****1.3.2 — Gauss's Law****1.3.3 — Applications of Gauss's Law**

TESTDAILY

1.3 Gauss's Law

1.3.1 Flux

1.3.1

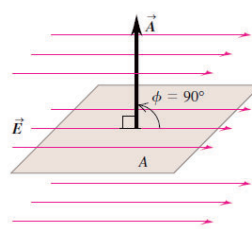
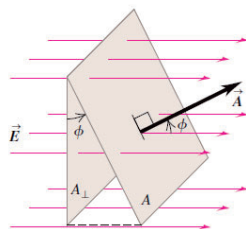
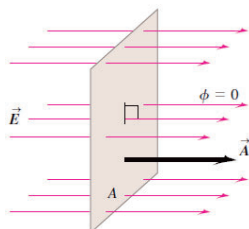
Definition

TESTDAILY

Electric Flux (Young 13e p729)

- A measure of the amount of electric field (or number of electric field lines) passing through the surface.
- Definition of electric flux for uniform field, flat surface

$$\Phi_E = EA \cos \theta = E_{\perp} A = \vec{E} \cdot \vec{A}$$



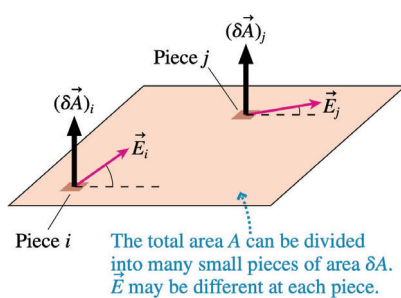
1.3.1

Definition

TESTDAILY

General Definition: Surface Integral

$$\Phi_E = \int E \cos \theta \, dA = \int E_{\perp} \, dA = \int \vec{E} \cdot d\vec{A}$$



Physical quantity	Electric flux 电通量
Definition	The scalar product of the electric field vector and the area vector over the entire area
Symbol	Φ_E
Equation	$\Phi_E = \int \vec{E} \cdot d\vec{A}$
Unit	$\text{N} \cdot \text{m}^2/\text{C}$

1.3.1

1.3 Gauss’s Law

1.3.2 Gauss’s Law



Theorem

TESTDAILY

Gauss’s Law (CED p38)

- The **total flux** through a **closed Gaussian surface** is proportional to the **charge enclosed** by the Gaussian surface.
- Integral form:

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

Note: the $\oint$ sign indicates surface integral over a closed surface.

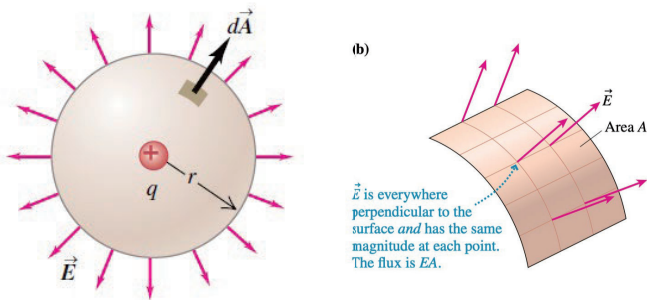


Example

TESTDAILY

Point Charge

Electric flux through a sphere centered on a point charge.

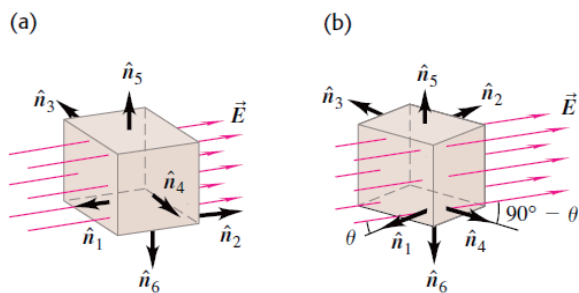


Example

TESTDAILY

Uniform Field

Electric flux of a uniform field $\vec{E}$ through a cubical box of side L in two orientations.



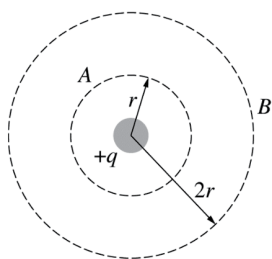
1.3.2

Exercises

TESTDAILY

2017_Intl_MCQ_3

A small sphere has a charge $+q$. Spherical Gaussian surfaces A and B are concentric with the sphere, as shown in the figure above. The radii of surfaces A and B are r and $2r$, respectively.



1.3.2

3. The magnitude of the electric flux through A is Φ_A . The magnitude of the electric flux through surface B is Φ_B . The ratio Φ_A/Φ_B is

- (A) 4/1
- (B) 2/1
- (C) 1/1
- (D) 1/2
- (E) 1/4

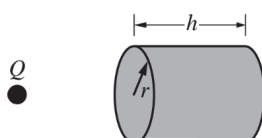
Exercises

TESTDAILY

2016_Intl_MCQ_26

A closed cylindrical shell of volume $\pi r^2 h$ is placed close to an object with a charge of Q , as shown in the figure above. There are no other charged objects nearby. The electric flux through the closed cylindrical shell is

- (A) 0
- (B) Q/ϵ_0
- (C) $\pi r^2 Q$
- (D) $2\pi r h Q$
- (E) $2\pi r^2 Q + 2\pi r h Q$



1.3.2

1.3 Gauss's Law

1.3.3 Applications of Gauss's Law

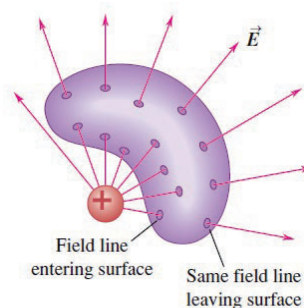
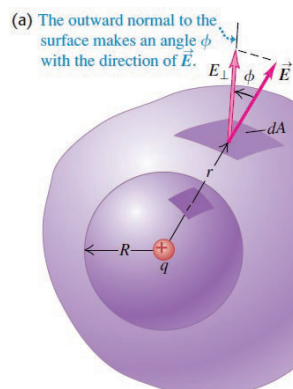
1.3.3

Concept

TESTDAILY

Gaussian Surface

- Gaussian surfaces are imaginary and can be chosen arbitrarily, but usually symmetrically to **simplify the expression of flux in terms of field.**



1.3.3

Concept

TESTDAILY

Symmetry

- We say that a charge distribution is symmetric if there is a group of geometric transformations that don't cause any physical change.
 - Translation
 - Rotation
 - Reflection



Original cylinder



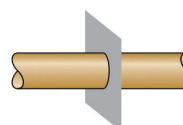
Translation parallel to the axis



Rotation about the axis



Reflection in plane containing the axis



Reflection perpendicular to the axis

1.3.3

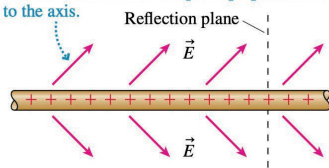
Concept

TESTDAILY

Symmetry

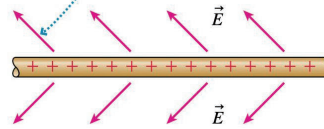
- The symmetry of the electric field must match the symmetry of the charge distribution.

(a) Is this a possible electric field of an infinitely long charged cylinder? Suppose the charge and the field are reflected in a plane perpendicular to the axis.

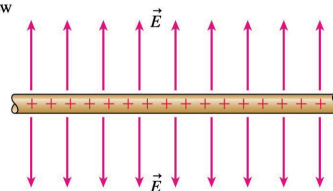


Reflect

(b) The charge distribution is not changed by the reflection, but the field is. This field doesn't match the symmetry of the cylinder, so the cylinder's field can't look like this.



Side view



1.3.3

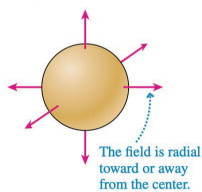
Method

TESTDAILY

Gaussian Surface

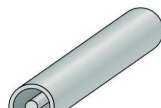
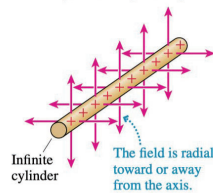
- Three types of symmetry

Spherical symmetry



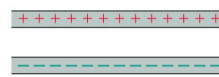
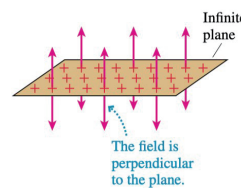
Concentric spheres

Cylindrical symmetry



Coaxial cylinders

Planar symmetry



Infinite parallel-plate capacitor

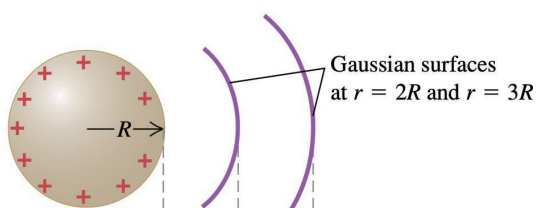
1.3.3

Example

TESTDAILY

Type I Spherical Symmetry: Example 1

Uniform charge distribution **on the surface** of a **conducting** sphere.

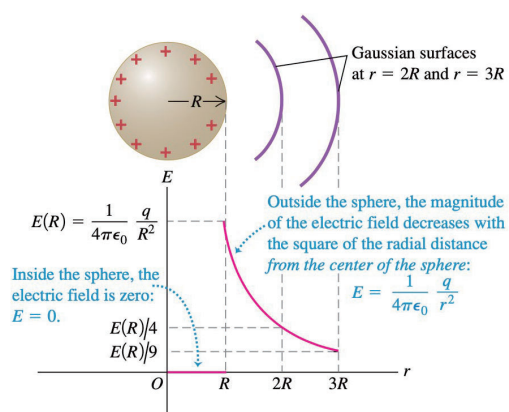


1.3.3

Example

TESTDAILY

Conducting Sphere



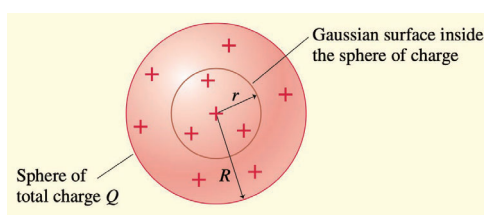
1.3.3

Example

TESTDAILY

Type I Spherical Symmetry: Example 2

- A **non-conducting** sphere with uniform charge density.

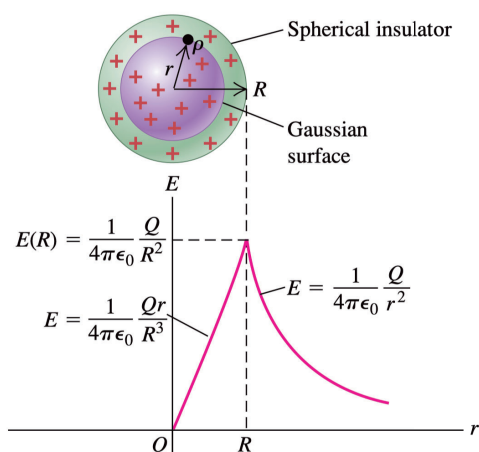


1.3.3

Example

TESTDAILY

Non-conducting Sphere



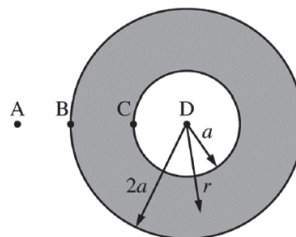
1.3.3

Example

TESTDAILY

Type I Spherical Symmetry: Example 3 2014_Intl_FRQ_3

A spherical insulating shell of radius $2a$ has a hollow cavity of radius a , as shown in the figure above. The charge density in the shell for $a < r < 2a$ varies according to the expression $\rho = br^2$, where b is a positive constant and r is the distance from the center of the shell. Express all algebraic answers to the following in terms of a , b , r , and fundamental constants, as appropriate.



1.3.3

Example

TESTDAILY

2014_Intl_FRQ_3

(a) Derive an expression for the total charge on the shell.

$$Q_t = \int \rho(r) dV$$

1.3.3

Example

TESTDAILY

2014_Intl_FRQ_3

(b) Using Gauss's law, derive an expression for the magnitude of the electric field in each of the following regions.

- i. $r > 2a$
- ii. $r < a$

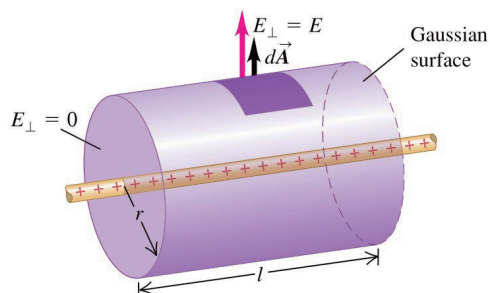
1.3.3

Example

TESTDAILY

Type II Cylindrical Symmetry: Example 4

A coaxial cylindrical Gaussian surface is used to find the electric field outside an **infinitely long, charged wire**.



1.3.3

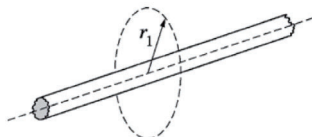
Exercises

TESTDAILY

2013_Intl_MCQ_33

An electric field is produced by the very long, uniformly charged rod shown above. If the strength of the electric field is E_1 at a distance r_1 from the axis of the rod, at what distance from the axis is the field strength $\frac{E_1}{10}$?

- (A) $\frac{r_1}{100}$
- (B) $\frac{r_1}{10}$
- (C) $\sqrt{10}r_1$
- (D) $10r_1$
- (E) $100r_1$



1.3.3

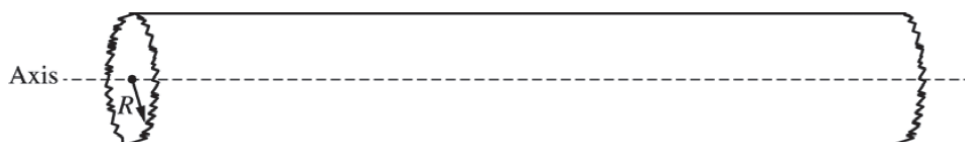
Example

TESTDAILY

Type II Cylindrical Symmetry: Example 5 2013_NA_FRQ_1

A very long, solid, nonconducting cylinder of radius R has a positive charge of uniform volume density ρ . A section of the cylinder far from its ends is shown in the diagram above. Let r represent the radial distance from the axis of the cylinder. Express all answers in terms of r , R , ρ , and fundamental constants, as appropriate.

- (a) Using Gauss's law, derive an expression for the magnitude of the electric field at a radius $r < R$. Draw an appropriate Gaussian surface on the diagram.



1.3.3

Example

TESTDAILY

2013_NA_FRQ_1



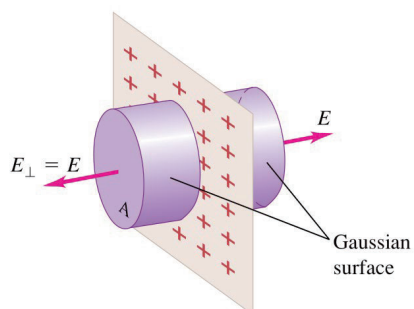
1.3.3

Example

TESTDAILY

Type III Planar Symmetry: Example 6

A cylindrical Gaussian surface is used to find the field of
infinite plane sheet of charge.



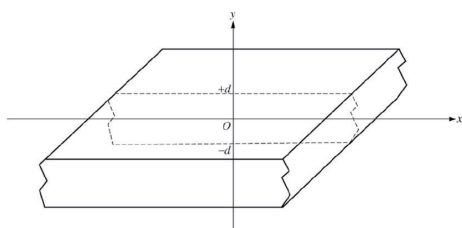
1.3.3

Example

TESTDAILY

Type III Planar Symmetry: Example 7 2013_Intl_FRQ_3

A nonconducting slab of infinite length and width and thickness $2d$ is portioned on a coordinate system as shown above. The slab is charged, and the charge per unit volume ρ is given by the expression $\rho(y) = C|y|$, where $-d \leq y \leq +d$ and C is a positive constant.



1.3.3

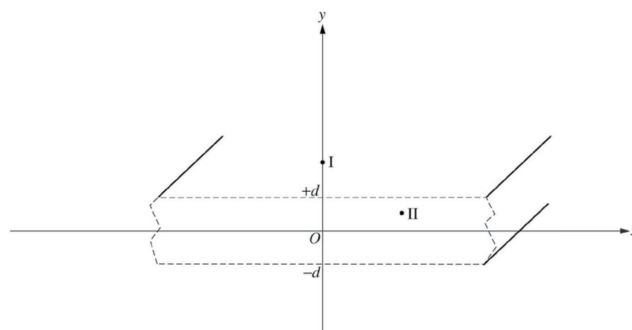
Example

TESTDAILY

2013_Intl_FRQ_3

(a) On the diagram below, do the following.

- Indicate with vectors the direction of the electric field at point I ($y > d$) and point II ($0 < y < d$).
- Draw a Gaussian surface that could be used to calculate the magnitude of the electric field at point I.



1.3.3

Example

TESTDAILY

(b) Using Gauss's law, derive expressions in terms of the given quantities and fundamental constants for the magnitude of the electric field E at the following points.

- Point I ($y > d$)

1.3.3

Example

TESTDAILY

(b) Using Gauss's law, derive expressions in terms of the given quantities and fundamental constants for the magnitude of the electric field E at the following points.

- Point II ($0 < y < d$)

1.3.3

Electric Potential

1.4



Electric Potential

1.4.1 — Electric Potential Energy

1.4.2 — Electric Potential

1.4.3 — Equipotential Surfaces

TESTDAILY

1.4 Electric Potential

1.4.1 Electric Potential Energy

1.4.1

TESTDAILY

Review

TESTDAILY

Conservative Force and Potential Energy

- If a force is conservative, the work done by the force is always the **opposite** of the change in potential energy U .

$$W_{a \rightarrow b} = -\Delta U$$

- The change in potential energy U is equal to the work done by external force to bring the objects from rest at one position to rest at another position.

$$W_{ext} = \Delta U$$

Review: Gravitational potential energy

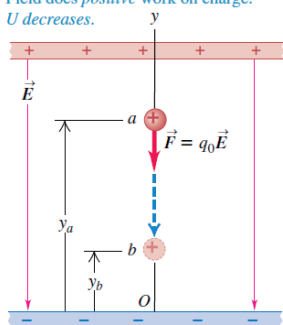


Introduction

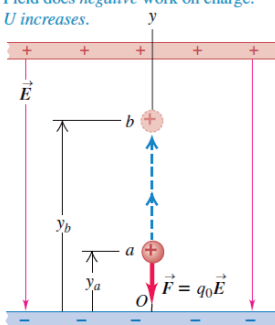
TESTDAILY

Electric Potential Energy in a Uniform Electric Field

- (a) Positive charge moves in the direction of $\vec{E}$:
- Field does *positive* work on charge.
 - U decreases.



- (b) Positive charge moves opposite $\vec{E}$:
- Field does *negative* work on charge.
 - U increases.



$$U_b = U_a + \Delta U_{a \rightarrow b} = U_a - q_0 E \Delta y$$

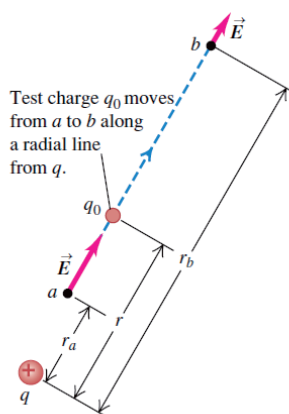


Property

TESTDAILY

Electric Potential Energy of Two Point Charges

$$U_b = U_a + \Delta U_{a \rightarrow b} = U_b + k \frac{qq_0}{r_b} - k \frac{qq_0}{r_a}$$

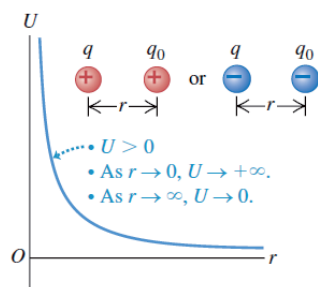
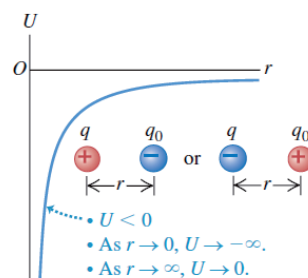


Property

TESTDAILY

Electric Potential Energy of Two Point Charges (CED p36)

$$U = k \frac{qq_0}{r}$$

(a) q and q_0 have the same sign.(b) q and q_0 have opposite signs.

1.4.1

Method

TESTDAILY

Conservation of Energy

Change in kinetic energy can be determined by using the principle of conservation of energy.

$$U_a + K_a = U_b + K_b$$

1.4.1

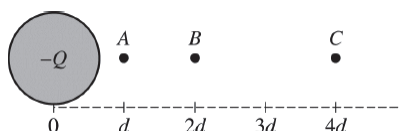
Exercises

TESTDAILY

2016_Intl_MCQ_34

A particle of charge $+q$ is released from rest at position B , which is a distance $2d$ from the center of a fixed nonconducting sphere that has a charge of $-Q$ distributed uniformly throughout its volume, as shown in the figure above. When the particle reaches position A , which is a distance d from the center of the charged sphere, its kinetic energy is K_0 . The same particle is now released from rest at position C , which is a distance $4d$ from the center of the nonconducting sphere. The kinetic energy of the particle when it again reaches position A is

- (A) K_0
 (B) $3K_0/2$
 (C) $2K_0$
 (D) $5K_0/2$
 (E) $3K_0$



1.4.1

Method

TESTDAILY

Conservation of Energy

- With external force

$$W_{\text{ext}} = \Delta E_{\text{sys}} = \Delta U + \Delta K$$

- When speed doesn't change

$$W_{\text{ext}} = \Delta U$$

1.4.1

Exercises

TESTDAILY

2016_Intl_MCQ_28

Two particles with positive charges q_1 and q_2 are both at rest and are very far apart. How much work would be done by external forces to bring the particles to rest at a distance d apart?

- (A) Zero
- (B) $\frac{kq_1q_2}{d}$
- (C) $-\frac{kq_1q_2}{d}$
- (D) $\frac{2kq_1q_2}{d}$
- (E) $-\frac{2kq_1q_2}{d}$

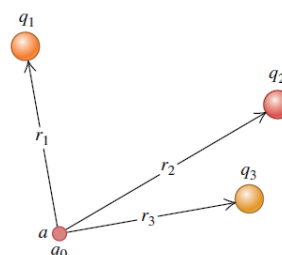
1.4.1

Definition

TESTDAILY

Potential Energy with Several Point Charges

For the entire system, U_{total} is the sum of the potential energies of interaction for each pair of charges



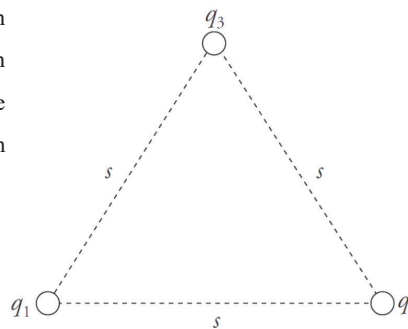
1.4.1

Example

TESTDAILY

Princeton_2020_Chapter13_Exmample6

Two positive charges, q_1 and q_2 , are held in the positions shown below. How much work would be required to bring (from infinity) a third positive charge, q_3 , and place it so that the three charges form the corners of an equilateral triangle of side length s ?



1.4.1

1.4 Electric Potential

1.4.2 Electric Potential

1.4.2

Definition

TESTDAILY

Electric Potential (Young 13e p761)

- We define the potential V at any point in an electric field as the potential energy U per unit charge **associated** with a test charge q_0 at that point.

Physical quantity	Electric potential 电势
Definition	The potential energy per unit charge
Symbol	V
Equation	$V = \frac{U}{q_0}$
Unit	volt, V 伏特 $1 \text{ V} = 1 \text{ J/C}$

1.4.2

Definition

TESTDAILY

Electric Potential of A Point Charge (CED p35)

- The electric potential at a particular location due to a single point charge is

$$V = k \frac{q}{r}$$

- Sign of potential
- Magnitude of potential



Method

TESTDAILY

Calculate Potential of Point Charges

Potential due to a collection of point charge

$$V = \frac{U}{q_0} = k \sum_i \frac{q_i}{r_i}$$

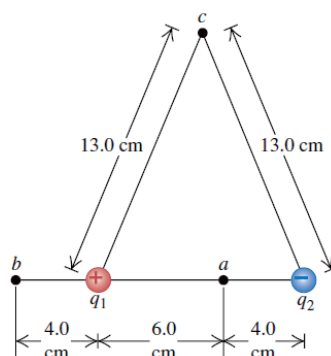


Example

TESTDAILY

Young_13e_p765_Example23.4

An electric dipole consists of point charges $q_1 = +12 \text{ nC}$ and $q_2 = -12 \text{ nC}$ placed 10.0 cm apart. Compute the electric potentials at points a , b , and c .



Answer: -900 V , 1930 V , 0



Method

TESTDAILY

Calculate Potential of Point Charges

Potential due to a continuous distribution of charge

$$V = k \int \frac{dq}{r}$$

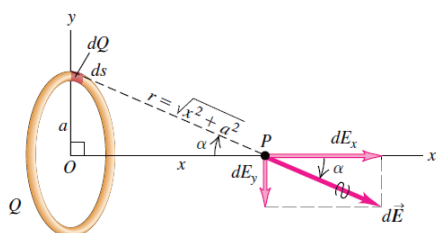
1.4.2

Example

TESTDAILY

Young_13e_p770_Example23.11

Calculate the electric potential along the central axis of the ring.

Answer: $k \frac{Q}{\sqrt{x^2 + a^2}}$

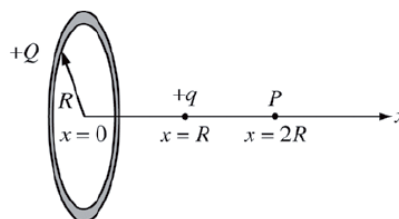
1.4.2

Exercises

TESTDAILY

2013_Intl_MCQ_24

A thin ring of radius R has charge $+Q$ distributed uniformly around the ring. The center of the ring is at the origin of an x -axis perpendicular to the plane of the ring, as shown in the figure above. A point charge $+q$ on the x -axis at position $x = R$ is released from rest. What is its kinetic energy when it reaches position P at $x = 2R$ on the x -axis?



(A) $\frac{3}{10} \frac{kQq}{R}$

(D) $\frac{kQq}{R}$

(B) $\frac{1}{2} \frac{kQq}{R}$

(E) $\frac{kQq}{R} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{5}} \right)$

(C) $\frac{1}{\sqrt{3}} \frac{kQq}{R}$

1.4.2

Method

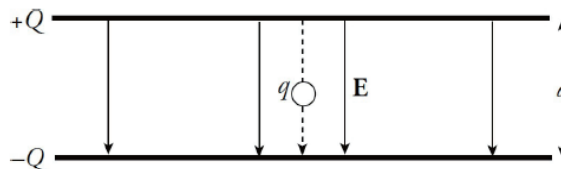
TESTDAILY

Calculate Potential Difference (Voltage 电压)

$$\Delta V_{a \rightarrow b} = V_b - V_a$$

Two large flat plates—one carrying a charge of $+Q$, the other $-Q$ —are separated by a distance d . The electric field between the plates, E , is uniform. Determine the potential difference between the plates.

$$\Delta V = \frac{\Delta U}{q}$$



1.4.2

Method

TESTDAILY

Relation between E & V

$$\Delta U_{a \rightarrow b} = -W_{a \rightarrow b} = -\int_a^b \vec{F} \cdot d\vec{r} = -\int_a^b q\vec{E} \cdot d\vec{r}$$

$$\Delta V_{a \rightarrow b} = \frac{\Delta U_{a \rightarrow b}}{q} = -\int_a^b \vec{E} \cdot d\vec{r} = -\int_a^b E \cdot d\vec{r}$$

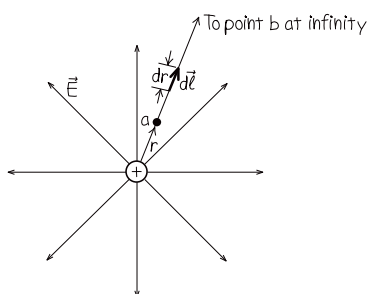
1.4.2

Example

TESTDAILY

Young_13e_p766_Example23.6

- Calculating the potential at a distance r from a point charge q by **integrating** the field for a single point charge.



1.4.2

Exercises

TESTDAILY

2016_Intl_MCQ_27

The electric field is zero everywhere within a certain region of space. The electric potential everywhere within the region

- (A) must be zero
- (B) must be uniform
- (C) must be positive
- (D) must be negative
- (E) can be both positive and negative in different parts of the region

1.4.2

Example

TESTDAILY

Young_13e_p769_Example23.10

Find the potential at a distance r from a very long line of charge with linear charge density (charge per unit length) λ .

$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{x}$$



1.4.2

Example

TESTDAILY

Princeton_2020_Chapter13_Example14

A nonconducting sphere of radius R has an excess charge of Q distributed uniformly throughout its volume (that is, the volume charge density, ρ , is a constant). The electric field at a distance $r < R$ from the center is given by the equation

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} r$$

What is the potential at a distance r from the sphere's center?

1.4.2

Method

TESTDAILY

Relation between E & V

- E_x is the negative of the derivative of V with respect to x

$$dV = -E_x dx$$

$$E_x = -\frac{dV}{dx}$$

- Alternate unit of electric field: volt per meter, V/m.

1.4.2

Exercise

TESTDAILY

2021_DE_D61102_5

The electric potential of a charge distribution as a function of position is given by the expression $V = 10x + 4x^2$, where V is in volts and x is in meters. What is the magnitude of the electric field at $x = 3$ m?

- A. 66 N/C B. 51 N/C C. 34 N/C D. 22 N/C E. 18 N/C

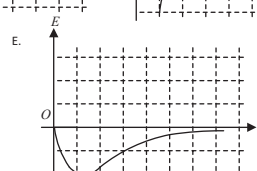
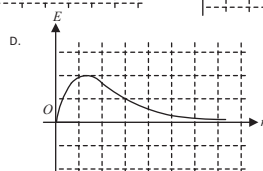
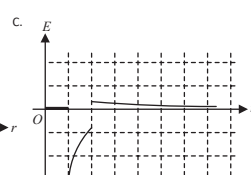
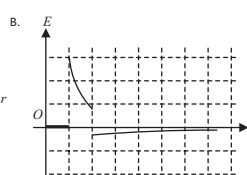
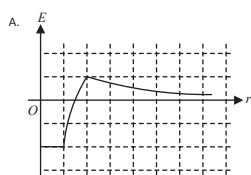
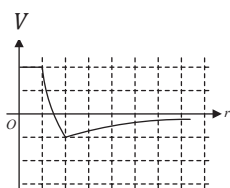
1.4.2

Exercise

TESTDAILY

2021_DE_D61102_8

A graph of electric potential V as a function of distance r from the center of spherically symmetric charge distribution is shown. Which of the following could represent the graph of electric field strength E as a function of distance r from the center of the same charge distribution?



1.4.2

1.4 Electric Potential

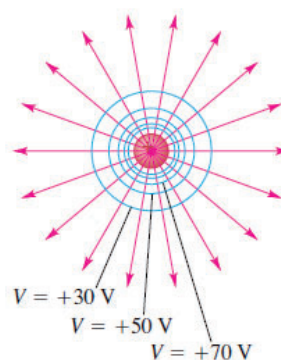
1.4.3 Equipotential Surfaces

1.4.3

Definition

Equipotential Surfaces (Young 13e p771)

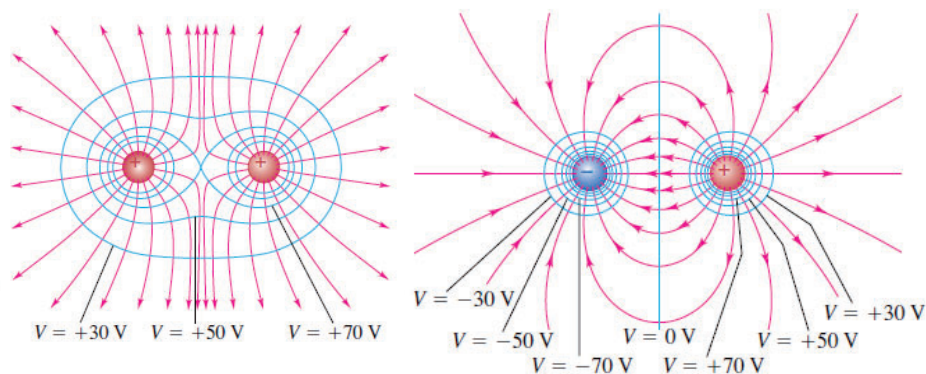
- An equipotential surface is a three-dimensional surface on which the electric potential is the same at every point. On a 2D cross-section plane, there are equipotential lines or isolines.
- If a test charge is moved from point to point on such a surface, the potential energy keeps constant, so the work done by electric force is zero.



1.4.3

Definition

Equipotential Surfaces



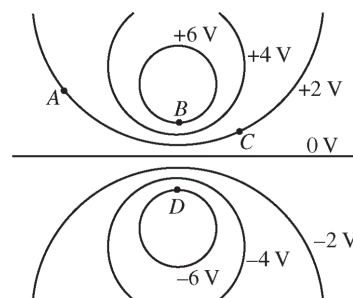
1.4.3

Exercise

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2012_Intl_MCQ_23

The diagram above shows a cross section of equipotential lines produced by a charge distribution. Points A , B , C , and D lie in the plane of the page.



23. For which two points can a negatively charged particle be moved from rest at one point to rest at the other with no work being done by the electric field?

- (A) A and B
- (B) A and C
- (C) A and D
- (D) B and C
- (E) B and D



Exercise

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2012_Intl_MCQ_24

24. A positively charged particle is moved by an external force from rest at one point to rest at another. For which of the following motions would net positive work be required by the external force?

- (A) From A to D
- (B) From B to A
- (C) From C to A
- (D) From C to D
- (E) From D to B



Exercise

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2012_Intl_MCQ_25

25. The electric potential shown in the diagram could be created by which of the following?

- (A) A ring of positive charge
- (B) A large sheet of positive charge
- (C) Two negative point charges
- (D) Two long lines of charge: one positive and one negative
- (E) A long line of positive charge and a negative point charge

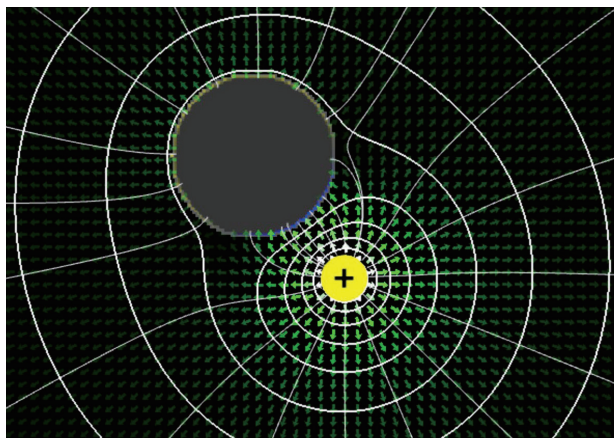


Exercise

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2012_Intl_MCQ_23~25

conducting cylinder and point charge



1.4.3

Property

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Equipotential Surfaces

- The direction of electric field is **perpendicular** to the equipotential line and pointing in the direction of **decreasing** potential.
- Electric field is greater where the equipotential lines are more closely together, smaller where the equipotential lines are farther apart.

1.4.3

Exercise

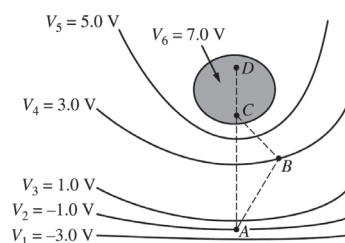
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2015_Intl_MCQ_30

Equipotential lines due to an electric field in a certain region of space are illustrated in the figure above. Points A and B are located on lines V_2 and V_4 , respectively, and points C and D are located within the equipotential region V_6 .

30. At which labeled point is the magnitude of the electric field the greatest?

- (A) A
 (B) B
 (C) C
 (D) D
 (E) It is the same at all the points.

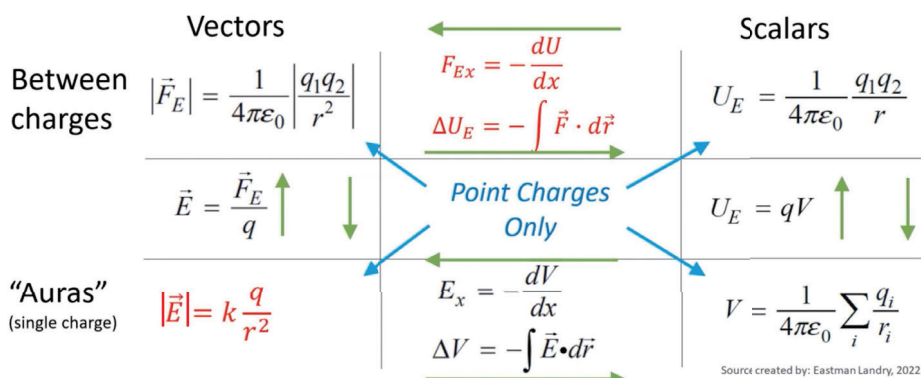


1.4.3

Summary

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Electrostatics Summary



1.4.3

Answer to MCQ

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Section	Source	Answer	Section	Source	Answer
1.1.2	2015_Intl_MCQ_10	C	1.4.1	2016_Intl_MCQ_28	B
1.1.4	2016_Intl_MCQ_35	C	1.4.2	2013_Intl_MCQ_24	E
1.2.3	2016_Intl_MCQ_10	C	1.4.2	2016_Intl_MCQ_27	B
1.3.2	2017_Intl_MCQ_3	C	1.4.2	2021_DE_D61102_5	C
1.3.2	2016_Intl_MCQ_26	A	1.4.2	2021_DE_D61102_8	B
1.3.3	2013_Intl_MCQ_33	D	1.4.3	2012_Intl_MCQ_23~25	B, E, D
1.4.1	2016_Intl_MCQ_34	B	1.4.3	2015_Intl_MCQ_30	A

1

Answer to FRQ

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2016_Intl_FRQ_1

(c) 3 points

Using a correct integral expression for Coulomb's law

$$E = \int \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} dq$$

For correctly expressing the charge element in terms of the linear charge density

1 point

$$\lambda_0 = Q/L \text{ so } dq = \lambda_0 dr$$

For correctly substituting into the first equation

1 point

$$E = \int \frac{1}{4\pi\epsilon_0} \frac{\lambda_0}{r^2} dr$$

For integrating with the correct limits or constant of integration

1 point

$$E = \frac{\lambda_0}{4\pi\epsilon_0} \int_{r=b}^{r=b+L} \frac{1}{r^2} dr = \frac{\lambda_0}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_{r=b}^{r=b+L} = \frac{\lambda_0}{4\pi\epsilon_0} \left(-\frac{1}{b+L} - \left(-\frac{1}{b} \right) \right)$$

$$E = \frac{\lambda_0}{4\pi\epsilon_0} \left(\frac{-b + b + L}{b(b+L)} \right)$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{\lambda_0 L}{b(b+L)}$$

1

Answer to FRQ

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2015_Intl_FRQ_3

15 points total

Distribution of points

(a) 4 points

For a correct expression for the y-component of the field element

$$dE_y = dE \cos \theta$$

1 point

For a correct expression for the field due to an element of charge on the wire

1 point

$$E = \frac{kq}{R^2}$$

$$dE = \frac{k dq}{R^2}$$

For expressing the charge element dq in terms of λ , R , and θ

1 point

$$dq = \lambda ds \text{ and } ds = R d\theta \text{ so } dq = \lambda R d\theta$$

For a correct substitution using the equations above

1 point

$$dE_y = dE \cos \theta = \left[\frac{k dq}{R^2} \right] \cos \theta = \frac{k (\lambda R d\theta)}{R^2} \cos \theta$$

$$dE_y = \frac{k \lambda \cos \theta}{R} d\theta = \frac{\lambda \cos \theta}{4\pi\epsilon_0 R} d\theta$$

1

Answer to FRQ

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2015_Intl_FRQ_3

(b) 5 points

For demonstrating that the x-component of the electric field is zero

1 point

$$\Sigma E_x = 0$$

For showing that the net field is due to the two concentric arcs

1 point

$$E = \Sigma E_y = E_{y\text{-inner}} + E_{y\text{-outer}}$$

For using $2R$ for the outer arc

1 point

$$E = \Sigma E_y = \int \frac{k \lambda \cos \theta}{R} d\theta + \int \frac{k \lambda \cos \theta}{2R} d\theta$$

For integrating with appropriate limits or constant of integration

1 point

$$E = \int_{-\pi/2}^{\pi/2} \frac{k \lambda \cos \theta}{R} d\theta + \int_{-\pi/2}^{\pi/2} \frac{k \lambda \cos \theta}{2R} d\theta$$

For a correct answer

1 point

$$E = \frac{3k\lambda}{R} = \frac{3}{4\pi\epsilon_0} \frac{\lambda}{R}$$

1

Answer to FRQ

TESTDAILY

2014_Intl_FRQ_3

15 points total

Distribution of points

(a) 5 points

For expressing the charge as an integral of the charge density

1 point

$$q = \int \rho dV$$

For a correct expression of the volume element

1 point

$$dV = 4\pi r^2 dr$$

For correctly substituting into the integral

1 point

$$q = \int br^2 (4\pi r^2) dr$$

$$q = 4\pi b \int r^4 dr$$

1

Answer to FRQ

TESTDAILY

2014_Intl_FRQ_3

For integrating with the correct limits of integration

1 point

$$q = 4\pi b \int_{r=a}^{r=2a} r^4 dr$$

$$q = 4\pi b \left[\frac{r^5}{5} \right]_{r=a}^{r=2a} = \frac{4\pi b}{5} (32a^5 - a^5)$$

For a correct answer

1 point

$$q = \frac{124}{5} \pi b a^5$$



Answer to FRQ

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2014_Intl_FRQ_3

(b)

i. 2 points

For a correct expression of Gauss's law

1 point

$$\frac{q}{\epsilon_0} = \oint E \cdot dA = E(4\pi r^2)$$

For correctly substituting the answer from part (a) into Gauss's law

1 point

$$\frac{124\pi b a^5}{5\epsilon_0} = E(4\pi r^2)$$

Correct answer

$$E = \frac{31ba^5}{5\epsilon_0 r^2}$$

ii. 1 point

For a correct answer
 $E = 0$

1 point



Answer to FRQ

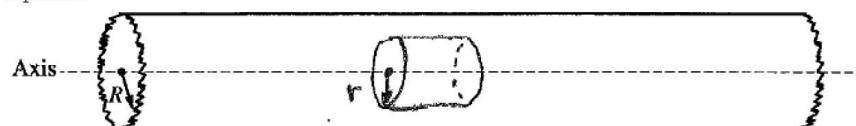
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2013_NA_FRQ_1

15 points total

Distribution of points

(a) 5 points



For drawing a Gaussian surface that is a cylinder, contained within and sharing the axis of the given cylinder

1 point

For using a correct statement of Gauss's law

1 point

$$\frac{q_{enc}}{\epsilon_0} = \oint E \cdot dA$$



Answer to FRQ

TESTDAILY

2013_NA_FRQ_1

For a correct expression for q_{enc} including the expression for the volume of a cylinder

1 point

$$q_{enc} = \rho V_{enc} = \rho(\pi r^2 L)$$

For a correct expression for the surface area of the sides of a cylinder

1 point

$$A = 2\pi r L$$

Substitute expressions into Gauss's law:

$$\frac{\rho(\pi r^2 L)}{\epsilon_0} = E(2\pi r L)$$

For a correct answer

1 point

$$E = \frac{\rho r}{2\epsilon_0}$$

1

Answer to FRQ

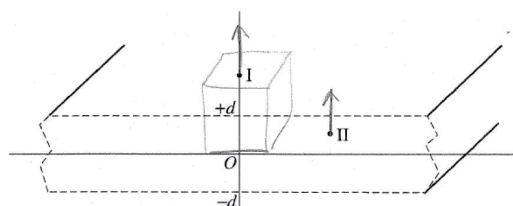
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Type III Planar Symmetry: Example 7 2013_Intl_FRQ_3

15 points total

Distribution of points

- (a)
i. 2 points



For drawing a vector toward the top of the page in the region $y > d$

1 point

For drawing a vector toward the top of the page in the region $0 < y < d$

1 point

The lengths of the vectors do not matter.

- ii. 4 points

For drawing any closed surface on the diagram

1 point

For a surface with the top and bottom parallel to the xz -plane

1 point

For a surface with sides parallel to the y -axis

1 point

For a surface with the bottom at either $y = 0$ or $y = -d$

1 point

1

Answer to FRQ

TESTDAILY

Type III Planar Symmetry: Example 7 2013_Intl_FRQ_3

- (b)
i. 3 points

For a correct expression of Gauss's law

1 point

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{enc}}{\epsilon_0}$$

For a correct expression for q_{enc}

1 point

$$q_{enc} = \int_0^d \rho dV = \int_0^d C|y|Ady = \frac{1}{2}CA[y^2]_0^d = \frac{1}{2}CA d^2$$

Substituting the expression for q_{enc} into Gauss's law

$$EA = \frac{\frac{1}{2}CA d^2}{\epsilon_0}$$

For a correct answer

1 point

$$E = \frac{Cd^2}{2\epsilon_0}$$

1

Answer to FRQ

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Type III Planar Symmetry: Example 7 2013_Intl_FRQ_3

(b) (continued)

ii. 2 points

For a correct expression for q_{enc} (must show correct limits)

1 point

$$q_{enc} = \int_0^y C|y|A dy = \frac{1}{2}CAy^2$$

For a correct answer

1 point

$$E = \frac{Cy^2}{2\epsilon_0}$$

Note: Full credit was awarded for replacing d with y in the answer for part (b)-i, even if the answer in part (b)-i is wrong.



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